

Temporal Anti-Aliasing for Single-Photon Reconstruction

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Single-photon avalanche diode (SPAD) cameras capture high-speed bursts of binary frames, where each pixel records only whether a photon arrived during a given exposure. A single frame carries almost no usable signal; a clean image only emerges from aligning and merging an entire burst (quanta burst photography). This project investigates single-photon image reconstruction under motion, specifically whether history-rejection heuristics from real-time temporal anti-aliasing (TAA) — neighborhood clamping and motion-adaptive blending — improve classical align-and-merge reconstruction, particularly under fast motion and disocclusion. We propose to evaluate on the Single Photon Challenge benchmark, co-organized by Prof. Gupta’s lab, comparing a TAA-augmented pipeline against naive-averaging and hierarchical block-alignment baselines.

1. Introduction

Single-photon avalanche diode (SPAD) cameras capture high-speed bursts of binary frames, where each pixel records only whether a photon arrived during a given exposure. A single frame carries almost no usable signal; a clean image only emerges from aligning and merging an entire burst, a process known as quanta burst photography. We address **single-photon image reconstruction under motion**: specifically, whether history-rejection heuristics from real-time temporal anti-aliasing (TAA) — neighborhood clamping and motion-adaptive blending — improve classical align-and-merge reconstruction, especially under fast motion and disocclusion.

The standard SPAD reconstruction pipeline (estimate motion, align, merge) is structurally identical to temporal anti-aliasing in real-time rendering, yet this connection does not appear to have been explored in prior work. Bringing history-rejection techniques developed for real-time graphics into photon-level reconstruction offers a genuinely cross-domain angle with a clear, testable hypothesis. SPAD arrays are also an emerging sensor technology — megapixel-scale arrays now exist — which makes the underlying reconstruction problem increasingly practical to study.

The project builds directly on Prof. Gupta’s own quanta burst photography research and the Single Photon Challenge benchmark his lab co-organized, giving the team a public dataset with ground truth rather than requiring custom data collection.

2. Related Work

Quanta burst photography (Gupta et al.) establishes the align-then-accumulate pipeline for SPAD bursts that serves as our baseline.

Temporal anti-aliasing. A survey of TAA techniques (Yang et al., NVIDIA, 2020) catalogs the history-rejection methods — neighborhood clamping, motion-adaptive blend weights — that we adapt to this domain.

Single Photon Challenge. This benchmark provides the

dataset and existing leaderboard submissions used for comparison in this work.

3. Approach

Our approach proceeds in four stages:

1. Implement a *baseline*: naive frame averaging, followed by hierarchical block alignment and merging (a simplified quanta burst photography pipeline).
2. Implement a *TAA-augmented pipeline* using the same alignment and merge backbone, adding neighborhood-clamping history rejection, motion-adaptive blend weights, and a per-pixel confidence map.
3. Evaluate both pipelines on held-out Single Photon Challenge training scenes and on synthetic motion sweeps generated with the visionsim simulator.
4. Analyze where and why each method breaks down under fast motion and disocclusion.

Figure 1 illustrates the per-frame mechanism of the TAA-augmented pipeline and the point at which the naive-averaging baseline diverges from it. This approach draws on course material including image alignment and transformations, RANSAC, optical flow, tracking, and the quanta computational imaging lecture.

4. Data

The Single Photon Challenge dataset provides 50 simulated training scenes and 5 test scenes (ground truth withheld), each burst (a “photon cube”) consisting of 1,024 binary frames. The full set is approximately 425 GB; we will work with a 5–10 chunk subset (roughly 8.5 GB per chunk), supplemented with our own synthetic motion sweeps generated via the MIT-licensed visionsim simulator.

5. Evaluation Plan

We will report PSNR and SSIM on held-out training scenes, and conduct a failure analysis against motion speed using visionsim sweeps to characterize when temporal reconstruction

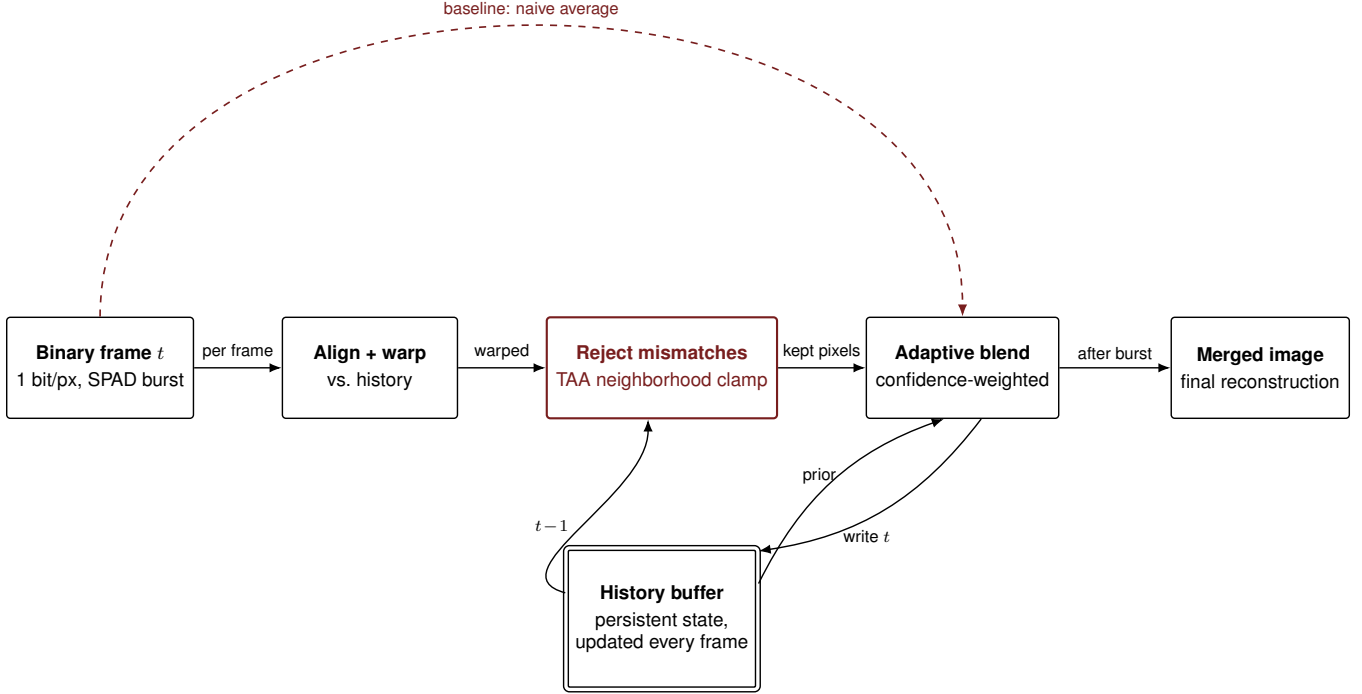


Figure 1: Per frame, the TAA-augmented pipeline (solid) warps the incoming binary frame to align with the running history buffer, rejects pixels where that alignment is unreliable, and adaptively blends the rest back into the history, which feeds the next frame and, after the full burst, the final reconstruction. The naive-averaging baseline (dashed) instead accumulates every frame directly, with no alignment or rejection.

breaks down. The contribution is framed as an interpretable classical method with a rigorous failure analysis, rather than a state-of-the-art claim against neural approaches.

6. Timeline

Wk	Dates	Milestone
1–2	Sep 29–Oct 12	Data pipeline and baseline working
3–4	Oct 13–Oct 26	TAA-augmented pipeline implemented
5	Oct 27–Oct 29	Mid-term report: baseline vs. TAA
6–8	Oct 30–Nov 19	Motion sweeps, failure analysis
9	Nov 20–Nov 30	Final evaluation, ablations
10	Dec 1–Dec 10	Presentation and final webpage

Table 1: Project timeline by course week.

7. Scope and Feasibility

This project topic was discussed with Prof. Gupta, who confirmed it is a reasonable and worthwhile direction to propose. As is expected at this stage, the specific scope may be refined as implementation progresses and initial results clarify what is achievable within the semester. Prof. Gupta also offered to connect the team with members of his lab for data access or implementation guidance if needed.